

Engineering Report

Phase 10 — UI Redesign, PWA, and Accessibility

SlotSense · Multi-tenant SaaS for community sports-facility booking

Project: sport-slot-dev · Region: asia-south1 · Repo: chandranakkalakunta/sport-slot-reservation

Executive summary

Phase 10 moved every page in the application off hand-rolled CSS onto a token-based design system (Tailwind v4 / shadcn/ui / Radix), delivered the app as an installable PWA with a working manifest and custom install flow, established a deliberate tenant-primary / platform-secondary co-branding hierarchy, and closed with a formal accessibility audit (28 automated scans, zero unresolved findings). Twenty-six pull requests shipped across nine sub-phases (10.1-10.9) plus a dedicated a11y pass (10.5), gated end-to-end by the CI/CD pipeline built in Phase 6 — every change tested, contract-checked against the tenant-theming boundary (ADR-0028), and merged with zero regressions reaching `main`.

The phase also surfaced and resolved a genuine production data-correctness bug — the AI booking agent and the My Bookings page disagreeing about a user's own bookings — diagnosed through a read-only investigation before any fix was attempted, and closed with a two-sided regression guard.

Outcomes

- **Design system adopted app-wide** — every page restyled onto Tailwind v4 / shadcn/ui / Radix primitives with real headings, consistent destructive-action posture, and light/dark theming that preserves per-tenant white-label branding.
- **PWA installability** — real app icon, valid manifest, working install flow independent of the browser's engagement heuristic, correct cache-freshness headers so deploys are visible without manual cache-clearing.
- **Co-branding hierarchy formalized** — tenant brand dominant in-header, SlotSense platform brand secondary via footer attribution, captured in ADR-0029 with explicit forward-binding consequences (install-prompt naming, notification sender identity, email branding).
- **List/card density standardized** — a single measured spec (content-height rows, consistent padding, inline actions at every breakpoint) applied uniformly, replacing per-page tuning.
- **Accessibility audit clean** — axe-core across 14 pages × 2 modes (light/dark), keyboard-only walkthrough of the full booking flow, Radix focus-trap behavior verified rather than assumed, booking-grid states confirmed accessible-name-driven rather than color-only.
- **A real booking-data bug found and fixed** — see Key Lessons.

Architecture: PWA cache-freshness split

The standard, durable pattern for SPA cache correctness — applied here for the first time in this project:

Layer	Cache-Control	Rationale
<code>index.html</code> , <code>manifest.webmanifest</code> , <code>sw.js</code> , <code>registerSW.js</code>	<code>no-cache</code>	Always revalidated; a deploy is visible on next load without manual intervention
<code>/assets/**</code> (content-hashed JS/CSS/fonts)	<code>public, max-age=31536000, immutable</code>	Filename changes when content changes, so permanent caching is safe and free

This is not a tradeoff between freshness and efficiency — the small "pointer" file always checks in cheaply (a `304` on no change); the large payload files are cached essentially forever.

Why the Hosting header deploy was unusual

SlotSense's frontend deploy does not use the standard `firebase deploy` CLI — Phase 6 built a custom REST API script (`deploy_hosting_rest.sh`) for the keyless-WIF auth path. The CLI is lenient about `firebase.json` shape and silently corrects loose formats; the REST API is not. Adding cache-control headers in Phase 10.7 surfaced two independent, sequential bugs in that stricter surface:

- **Deploy rejected outright.** The headers config was written in a shape (`{ "Cache-Control": "no-cache" }`), a map the REST API does not accept — it requires a **list of** `{key, value}` objects per rule. Every deploy carrying this config failed at `POST .../versions` with `HTTP 400: Cannot bind a list to map for field 'headers'`. The frontend

- never shipped; `main` looked merged, the live site was not updated.
- **Deploy succeeded, header still wrong.** After correcting the JSON shape, deploys succeeded — but a `curl` with a cache-busting query (confirmed `x-cache: MISS`, i.e. reaching origin, not a CDN copy) still showed the old header. Firebase Hosting matches header rules against the **requested path**; a rule scoped to `/index.html` does not apply to a request for `/`, even though Hosting serves `index.html`'s content there via rewrite. An explicit `source: "/"` rule was required in addition.

Convergence trail — cache header deploy

Stage	Signal	Fix
Deploy	<code>400 INVALID_ARGUMENT</code> — "Cannot bind a list to map"	Corrected headers JSON to <code>{source, headers: [{key,value}]}</code> shape
Verification	<code>cache-control: max-age=3600</code> persists after successful deploy, <code>x-cache: MISS</code> (origin, not CDN)	Added explicit <code>source: "/"</code> rule alongside <code>/index.html</code>
Result	<code>cache-control: no-cache</code> , <code>x-cache: MISS</code>	Confirmed live

Each signal precisely named the next fix — the same diagnostic signature as Phase 6's WIF convergence trail, on a different surface of the same deploy pipeline.

Convergence trail — booking data-correctness bug

Surfaced by the Coordinator's own testing: the AI agent reported 3 confirmed bookings for a user; the My Bookings page reported zero; the agent's own cancel tool reported nothing cancellable within its window — three views of the same data, disagreeing. Rather than patch the symptom, a read-only diagnostic was run first.

Path	Behavior found	Root cause
My Bookings page	Returns 0	Repo orders by deterministic doc-ID (<code>facility/date/start</code>); page fetches <code>limit=20</code> ; ≥ 20 accumulated docs (incl. cancelled) push today's confirmed bookings past the fetch window
Agent "my bookings"	Returns 3	Same query, <code>limit=100</code> — happens to see the same records the page misses
Agent "cancel"	Finds nothing to cancel	Sport-lookup built from <i>active-only</i> facilities; a deactivated court's booking hits <code>fac = None</code> and is silently skipped

Fix: moved "upcoming" filtering to the backend as a proper query filter (`date >= from_date`) computed once, shared by both page and agent — instead of client-side filtering over a partial, paginated result. Cancel's facility lookup now spans active *and* inactive facilities for sport-name resolution, while the agent's display list stays active-only. Both fixes were shipped with tests proving the false-negative was resolved **and** that no false-positive was introduced (a genuinely non-cancellable booking still can't be cancelled).

Key lessons

Layout-geometry bugs masquerade as spacing bugs

The single largest time cost of the phase was card/list density (~8 sub-phase rounds, 10.6a–10.6i). Two padding-focused fixes were tried and were each partially right but not the actual cause; the real cause — a CSS grid's `align-items: stretch` acting on a single-child flex column — was layout geometry, not spacing, and was found only once actual DevTools/Playwright measurement replaced reasoning from class names. **Standing rule going forward:** if an element is taller than its content and padding is already zero, suspect stretch/min-height/grid before touching spacing values again.

"Merged" is not "deployed" is not "deployed correctly"

Two deploy-pipeline bugs (above) were each invisible from the GitHub merge UI — one silently failed the actual Hosting release, the other released successfully but with the wrong runtime behavior. **Standing rule:** after any merge to `main`, confirm the `Deploy (main)` run succeeded (`gh run list --workflow="Deploy (main)"`) and independently verify the live behavior (cache-busted `curl`, not just "the merge went through").

Tailwind v4's spacing scale does not match prior assumptions

`p-4` measured 32px in this project's configuration, not the commonly-assumed 16px — discovered only via direct pixel measurement partway through the density investigation. Verify a framework's actual computed output early; don't reason from class names against a remembered scale.

Diagnose before fixing, especially the second time a symptom recurs

The booking bug was solved cleanly on the first attempt because a read-only investigation preceded any code change. Several visual bugs earlier in the phase (card height, back-link visibility) cost multiple rounds precisely because a plausible-sounding cause was patched before being measured. This is now standing practice for any bug that has already survived one fix attempt.

Operational notes & follow-ups

- ADR-0029 (PWA Co-Branding Hierarchy) committed alongside this report.
- Playwright added as a standing devDependency — available for future visual/layout diagnostics without ad-hoc setup.
- Custom install-prompt banner (`InstallPrompt.tsx`) does not wait on the browser's `beforeinstallprompt` engagement heuristic; falls back to explicit platform instructions ("your browser's ⋮ menu") when the native event hasn't fired.
- Deferred: a fully independent numeric dark-mode contrast re-check (axe-core via jsdom cannot compute CSS custom-property values; the one known dark-mode contrast issue — back-link FOUC — is fixed and verified, but a dedicated dark-mode Lighthouse pass would close the last gap).
- Phase 11 begins the feature roadmap: non-continuous facility slot openings, ahead of multi-capacity slots and billing (both explicitly deprioritized), per Coordinator direction.

Status: Phase 10 complete. Design system, PWA, co-branding, and accessibility are the enforced state of the application; the booking data-correctness fix is live and verified in production.